BENTONVILLE, ARKANSAS

KARST

Architecture Brief

407–409 SE F St — SE Market District

February 2026

Confidential

Architecture Brief — Bentonville Wellness Center

> *Detailed textual description optimized for AI architecture and rendering tools. Use this document as the primary input for concept renders, 3D visualization, and architectural AI generation.*

Building Overview

Type: Adaptive reuse of distribution warehouse + new construction **Address:** 407-409 SE F St, Bentonville, AR 72712 **Site context:** Adjacent to Ice House building, near The Momentary (Crystal Bridges), on Razorback Greenway trail **Zoning:** DN-4 mixed-use **Estimated total area:** 30,000-40,000+ sq ft across all levels **Levels:** Underground + Ground + Upper + Rooftop **Structure:** Multi-story — new structure within/around existing warehouse shell

Architectural Style: "Ozark-Nordic"

A fusion of three traditions applied to a single building:

E. Fay Jones Organic Architecture

- Steep gables and angular roof forms that echo Ozark ridgelines
- Deep integration with landscape — building appears to grow from the site
- Local stone (Ozark limestone) as primary structural/facing material
- Exposed timber structure — mass timber beams, columns, trusses
- Narrow, vertical windows that frame landscape views
- Reference: Thorncrown Chapel (Eureka Springs, AR) — soaring timber, glass, stone, forest integration

Nordic Warm Minimalism

- Cedar paneling on interior walls and sauna areas — warm, aromatic, natural
- Clean lines — minimal ornamentation, let materials speak

- Atmospheric lighting — warm color temperature (2700-3000K), indirect, designed for mood
- Sauna/bathhouse tradition — communal, egalitarian, contemplative
- Reference: Löyly Helsinki (Avanto Architects) — slatted wood exterior, sauna culture, waterfront

Japanese Onsen Sensibility

- Reverence for water, stone, and transition between spaces
- Material honesty — no applied finishes hiding structure
- Careful threshold design — each transition (dry to wet, light to dark, hot to cold) is intentional
- Acoustic design — water sounds, silence, natural materials absorbing harsh frequencies
- Reference: Kengo Kuma — layered timber, diffused light, landscape integration

Industrial Heritage

- Preserve exposed structural elements from original warehouse where architecturally interesting
- Steel beams, columns, connections left visible — contrast with warm organic materials
- Original warehouse roof height (~25-30ft) used for climbing wall and dramatic vertical space
- Patina and age of warehouse steel celebrated, not hidden

Level-by-Level Program

Underground — “The Cave”

Purpose: Contrast therapy core. The deepest, most grounded space. Primal. Internal. Silent.

Spatial requirements: - Excavated below grade — minimum 10ft ceiling height, 12ft preferred for communal sauna - Total underground area: 6,000-8,000 sq ft - Must accommodate significant mechanical infrastructure (water heating, filtration, ventilation, humidity control)

Rooms and features:

ROOM	DIMENSIONS	KEY FEATURES
Communal sauna	600-800 sq ft, 12ft ceiling	Multi-tier cedar seating (3 tiers), large format. Seating for 40+ persons. Traditional Finnish dry sauna, 180-200°F. Cedar-paneled walls and ceiling. Atmospheric downlighting embedded in ceiling. Large stones/boulders as heat mass. No windows — completely interior.
Cold plunges	400-500 sq ft total	3-4 plunge pools at different temperatures (45°F and 55°F). Natural stone or concrete basin construction. Riverstone surround. Overhead lighting mimicking light filtering through cave ceiling.
Steam room	300-400 sq ft	Eucalyptus-infused steam. Stone-tile walls. Built-in stone seating. 100% humidity, 110-115°F. Minimal visibility — atmospheric.
Mineral baths	500-600 sq ft	2 warm-water soak pools (95-102°F). Natural stone construction. Variable depth (sitting and standing areas). Subtle underwater lighting.
Wet rooms / showers	800-1,000 sq ft	20+ shower stations. Mix of rain showers and handheld. Stone/tile flooring with drainage. Private and open options.
Changing facilities	600-800 sq ft	Full-size lockers (100+). Grooming stations. Vanity area. Separate men's/women's with potential unisex area.
Transition corridors	As needed	Narrow stone corridors connecting spaces. Riverstone floors. Dim path lighting at ankle level. Sound of water throughout.

Materials palette: - **Walls:** Rough-cut Ozark limestone blocks (natural face, not polished). Some areas with cedar paneling. - **Floors:** Riverstone mosaic in wet areas. Slate or limestone in dry circulation areas. - **Ceiling:** Exposed stone/concrete with embedded indirect lighting. No dropped ceiling tiles. - **Fixtures:** Matte black or dark bronze hardware. No chrome. - **Lighting:** Warm (2700K), indirect, dimmable. Recessed cove lighting. Some candle/flame elements in relaxation areas.

Atmosphere targets: - Sound: Water flowing, occasional steam release, silence. Acoustic dampening on all surfaces. - Light: Very low. Eyes should adjust upon entering. Light sources should feel like firelight or filtered cave light. - Temperature: Varies dramatically by room (38°F to 200°F). Transition

corridors neutral (68-72°F). - Scent: Cedar, eucalyptus, wet stone. No artificial fragrances. - Cell-phone-free zone — no screens, no devices.

Design references: - Blue Lagoon Retreat, Iceland (Basalt Architects) — volcanic stone, atmospheric pools, minimal lighting - Othership, Toronto — cedar interiors, riverstone, breathwork-designed sauna - Therme Vals, Switzerland (Peter Zumthor) — monolithic stone, water, light slots - Juvet Landscape Hotel, Norway (JSA) — minimal intervention in landscape

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Ground Level — “The Foundation”

Purpose: Movement, training, nutrition, community entry. Where the public first encounters the building.

Spatial requirements: - Primary entrance at grade on SE F St - Secondary entrance from garden courtyard / Greenway trail side - Total ground level area: 10,000-14,000 sq ft - Full warehouse ceiling height (~25-30ft) in main strength area for climbing wall - Studios can be lower ceiling (10-12ft)

Rooms and features:

ROOM	DIMENSIONS	KEY FEATURES
Reception / gathering	800-1,000 sq ft	Welcome desk. Community seating. Display of daily programming. Warm materials — timber, stone. Not corporate — more lodge/home feeling.
Strength floor	3,000-4,000 sq ft	Premium equipment (Eleiko, Rogue, or similar) integrated into architecture. Equipment placement follows spatial design, not random grid. Mirrors minimal — prefer views/glass. Full warehouse ceiling height. Natural light from clerestory or upper-level glass. Rubber flooring with wood inlays.
Functional movement / martial arts mat	1,500-2,000 sq ft	Open flexible space. Sprung floor. Can host martial arts, Ido Portal movement, primal flow, mobility classes. Retractable/storable equipment. Adjacent to strength floor.
Studio rooms (2-3)	800-1,000 sq ft each	Configurable: hot yoga, mat Pilates, breathwork, group classes. Good ventilation. Acoustic isolation. Mirrors optional per room. Lighting programmable (bright for Pilates, dim for breathwork).
Hot studio (1)	600-800 sq ft	Enclosed, sealed for heat (95-105°F). Very dim lighting. Intimate feeling — max 20-25 persons. Infrared heat panels + traditional heat. Dark materials. Reference: Yoga Tropics aesthetic — womb-like, immersive.
Nutrition bar	500-700 sq ft	Counter service + 8-12 seats. Cold-pressed juice, smoothies, prepared foods. CSA popup display (local farm produce). Fresh ingredient storage. Visible from street — public-facing storefront.
Rock climbing wall	800 sq ft floor, 25-30ft height	Uses original warehouse vertical space. Textured climbing surface — natural stone aesthetic, not plastic-gym look. Auto-belays. Top-rope stations. Bouldering area at base. Dramatic visual feature visible from entrance. LED accent lighting on wall face.

Materials palette: - **Floors:** Engineered hardwood or wood-look in reception. Rubber with wood inlays on strength floor. Sprung wood in studios. Stone tile at entry and nutrition bar. - **Walls:** Mass

timber paneling. Exposed original warehouse steel where interesting. Some limestone accent walls. - **Ceiling:** Exposed timber structure + original warehouse steel trusses. Pendant and track lighting. - **Equipment finish:** Matte black equipment, wood accents. No neon, no chrome commercial gym aesthetic.

Atmosphere targets: - Sound: Energetic but controlled. Acoustic panels in studios. Music in strength area. Natural sounds in circulation. - Light: Abundant natural light from clerestory/upper windows and courtyard-facing glass. Warm artificial supplements. - Temperature: Comfortable (68-72°F) in general areas. Hot studio sealed and separately controlled.

Design references: - Equinox flagship clubs — premium equipment integration, material quality - John Pawson — minimal, warm, restrained material palette - The Record hotel (Bentonville) — local boutique hospitality material language

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Upper Level(s) — "The Canopy"

Purpose: Mind, media, learning, community gathering. The intellectual and creative layer.

Spatial requirements: - Accessed by feature staircase and elevator - Total upper level area: 6,000-8,000 sq ft - Higher glass-to-wall ratio than lower levels — greenhouse effect - Structural capacity for living plant walls and planter systems

Rooms and features:

ROOM	DIMENSIONS	KEY FEATURES
Media production studio	600-800 sq ft	Sound-isolated (STC 55+). Podcast setup (2-4 mic positions). Video-capable (3+ camera angles). The space IS the set — visible through glass wall to library. Designed to be visually stunning on camera. Acoustic panels integrated as design feature. Professional lighting rig (dimmable, color-adjustable). Cable management and tech infrastructure built in.
Library / sound lounge	1,500-2,000 sq ft	Curated health, wellbeing, philosophy, science collection (500+ volumes). Reading nooks with upholstered seating. Chess tables (2-3). Tabletop game storage. Sound lounge zone with premium headphones/listening stations. Quiet zone — no calls, soft conversation only. Living plant wall as backdrop.
Meeting / event space	1,000-1,500 sq ft	Configurable: talks, lectures, workshops, community dinners. Seating for 40-60. A/V system for presentations. Retractable partition to expand into library space for larger events.
Cowork nooks	400-600 sq ft total	3-4 small zones with power, WiFi, comfortable seating. Not a full coworking space — informal, no reservations.
Exam / PT rooms (2)	200-250 sq ft each	Small enclosed rooms for practitioner visits. Treatment table. Sink. Storage. Sound-isolated.

Materials palette: - **Walls:** Light timber paneling. Large glass panels. Living plant walls (green wall system with integrated irrigation). - **Floors:** Light hardwood or polished concrete. Area rugs in library/lounge zones. - **Ceiling:** Exposed timber beams. Skylights or clerestory glass. Pendant lighting — warmer, residential scale. - **Furniture:** Warm woods, natural fabrics, leather. Scandinavian-Japanese fusion. No office/commercial furniture.

Atmosphere targets: - Sound: Quiet. Acoustic design priority. Library and media studio need excellent sound isolation. Natural sounds (plants, water features if any). - Light: Maximum natural light. Greenhouse/conservatory feeling. Glass walls facing south/west for afternoon light. Operable shades for media studio light control. - Temperature: Comfortable year-round. Extra ventilation for plant walls. - Feeling: Like being in a treehouse or greenhouse. Elevated, light, alive with plants.

Design references: - Aesop stores — curated material palette, library sensibility - Kengo Kuma — layered timber, filtered light, green integration - Snohetta libraries — warm, civic, light-filled - Studioilse — warm Scandinavian interiors, intellectual but comfortable

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Rooftop — “The Sky”

Purpose: Outdoor recovery, views, garden, movement. The culmination of the vertical journey.

Spatial requirements: - Structural capacity for soil/garden loads, gathering, and furniture - Total rooftop area: 4,000-6,000 sq ft usable - Perimeter safety (railings integrated into design — not bolt-on industrial) - Weather protection zones (pergola/shade structures)

Features:

FEATURE	DESCRIPTION
Outdoor lounge	Recovery seating (daybeds, loungers, chairs). Shade structures — timber pergola with retractable fabric. Views to Ozark hills and downtown Bentonville skyline. Fire features for cool weather.
Rooftop garden	Edible and ornamental plantings in raised timber beds. Herbs and ingredients feeding nutrition bar. Echoes Ice House’s planned rooftop garden next door.
Grass strip	Real grass surface (or premium synthetic) for outdoor yoga, stretching, barefoot movement. 15-20 person capacity.
Planting / landscape	Native Ozark species. Drought-tolerant. Pollinator-friendly. Seasonal interest. Low maintenance.

Materials palette: - **Decking:** Thermally-modified timber or quality composite. Warm tone. - **Railings:** Minimal cable or glass — maximize views. - **Furniture:** Teak or similar outdoor hardwood. Matte finishes. - **Planters:** Corten steel or raw timber.

Design references: - BIG (Bjarke Ingels Group) — walkable green roofs, landscape as architecture - Snohetta — civic rooftop spaces, landscape extensions - High Line (NYC) — elevated public landscape, native planting, integrated seating

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Exterior

Features:

FEATURE	DESCRIPTION
Art court	Basketball/pickleball convertible court. Custom art design on court surface (reference: art courts in NWA/Fayetteville). Public-facing. Lit for evening play.
Bike parking	30+ bike rack capacity. Covered section. Direct Greenway trail access. Bike repair station.
Garden courtyard	Connecting building to Greenway trail and The Momentary green space. Native Ozark plantings. Seating. Water feature (optional). Gathering space for outdoor community events.
Nutrition co-op storefront	Street-facing entrance separate from main membership entrance. Awning/canopy. Outdoor counter seating.
Stormwater management	Permeable paving. Rain gardens. Bioswales. Integrated into landscape design.

Sustainability Strategy

High-performance sustainability integrated into design — not pursuing formal certification (LEED, etc.) but exceeding typical standards:

SYSTEM	APPROACH
Energy — solar	Rooftop photovoltaic array sized to offset significant portion of building load
HVAC — geothermal	Ground-source heat pumps where feasible — particularly efficient for contrast therapy (heating sauna, cooling plunges)
Structure — mass timber	Cross-laminated timber (CLT) or glulam primary structure — low embodied carbon, biophilic
Envelope — materials	Local Ozark limestone (low transport emissions), regional timber sourcing
Water — rainwater	Collection system feeding landscape irrigation and potential grey-water reuse
Water — thermal	Heat recovery from sauna/steam water systems to preheat incoming water
Biophilia — living walls	Living plant walls for air quality, humidity regulation, and occupant wellbeing
Biophilia — rooftop garden	Stormwater retention, urban heat island mitigation, edible production
Lighting	LED throughout + natural light optimization (skylights, clerestory, glass). Daylight harvesting controls.
Ventilation	Natural ventilation design in upper levels and rooftop. Mechanical ventilation with heat recovery in sealed areas.

Technology Integration

Technology is embedded in infrastructure, never in aesthetic. The building should feel analog and natural — technology works invisibly.

SYSTEM	DESCRIPTION
Wearable integration	Oura, Whoop, Apple Watch data syncs with personal dashboard. Recovery recommendations based on HRV, sleep, readiness.
Climate control	Smart zone-by-zone climate management — critical for spaces ranging from 38°F (plunge) to 200°F (sauna). Automated humidity control.
Biometric check-in	Fingerprint or NFC-based member access. No key cards.
Content infrastructure	Camera-friendly lighting throughout (high CRI, dimmable). Acoustic design for filming in multiple locations. Dedicated fiber/ethernet to media studio.
Booking / scheduling	Digital class booking, sauna session timing, practitioner appointments — all via member app.
Water quality	Continuous monitoring of all pool/plunge/bath water chemistry. Automated treatment systems.

Rendering Guidance for AI Tools

When generating concept renders, prioritize these visual qualities:

Exterior shots: - Dusk/golden hour lighting showing warm interior glow through glass - Razorback Greenway trail visible with cyclists/pedestrians - Ozark limestone facade with mass timber accents - Rooftop garden visible from street level - The Momentary's form visible in background for site context - Art court with dynamic court surface art - Garden courtyard with native plantings

Underground (The Cave): - Dim, warm atmospheric lighting — like firelight in a cave - Steam/mist in air creating depth - Cedar-lined sauna with multi-tier seating, bodies in silhouette - Cold plunge pools with still water reflecting dim light off stone walls - Riverstone floors with water - No visible technology — timeless, could be ancient or future

Ground Level (The Foundation): - Climbing wall as dramatic vertical feature — natural stone aesthetic, figures climbing - Natural light filtering from clerestory above - Timber and stone — warm, minimal, not commercial gym - Nutrition bar with fresh produce display - Original warehouse steel structure visible alongside new timber

Upper Level (The Canopy): - Greenhouse atmosphere — abundant plants, glass, filtered light - Library with curated books, reading nooks, warm lighting - Media studio visible through glass —

intimate, designed space - Living plant walls — lush, varied, textured - Warm timber + greenery — treehouse feeling

Rooftop (The Sky): - Panoramic Ozark hills view - Outdoor yoga on grass strip — figures in poses, morning light - Rooftop garden with raised timber beds - Fire feature and lounge seating for evening atmosphere - Connection to sky — expansive, free, open

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This brief is designed as the primary input for architectural AI tools (Midjourney, DALL-E, Stable Diffusion, AI architecture generators). Each section provides sufficient detail for prompt engineering while maintaining design coherence.

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